

Aileron Direct-Drive Pin Load Calculation

Preliminary aerodynamic hinge-moment, direct-drive torque, and pin sizing calculation

Purpose. This document calculates the loads that must be known before selecting a servo for the shown direct-drive aileron concept. It does not select a servo. The calculation covers the aerodynamic aileron normal force, hinge moment, direct-drive torque, rolling moment, pin torsional stress, pin twist, and practical minimum pin dimensions.

Important limitation. The exact aileron hinge moment requires the hinge-moment coefficient for the specific airfoil, aileron chord ratio, hinge gap, hinge line, Reynolds number, and deflection. Since that coefficient is not available here, the calculation uses a conservative preliminary method based on aileron normal force and a center-of-pressure arm. This is suitable for first mechanical sizing, but final sizing should be checked by airfoil/control-surface data, CFD, wind-tunnel data, or a physical load test.

Source basis. The Fundamentals of Flight Control lecture slides identify the aileron as the primary roll-control surface and show the actuator role in the flight-control chain. Scholz's actuator-sizing note gives the hinge-moment form $HM = c_h q A c$ and explains that hinge moments are used to derive actuator force levels. The same logic is applied here to an electric direct-drive servo: calculate maximum hinge moment first, then apply efficiency and safety factor to obtain the required direct-drive torque.

1. Input values and assumptions

Property	Symbol	Value	Unit	Note
Air density	ρ	1.17	kg/m ³	From current design table
Cruise / design speed used here	V	24	m/s	Use V_{max} instead if the aircraft can fly faster
Stall speed, comparison only	V_{stall}	14	m/s	Not used for final torque sizing
Mean wing chord	c	0.300	m	Used for Reynolds number
Kinematic viscosity	ν	1.44e-05	m ² /s	From current design table
Single aileron span	b_a	0.7568	m	From current design table
Single aileron chord	c_a	0.075	m	25% of 0.30 m wing chord
Aileron deflection cases	δ_a	10, 15, 20	deg	20 deg is used as conservative design case
Aerodynamic effectiveness factor	τ	0.5 to 1.0	-	1.0 is used as conservative value
Center-of-pressure arm behind hinge	x_{cp}	$0.4 c_a = 0.030$	m	Preliminary assumption
Safety factor	SF	3	-	Applied to direct-drive torque
Drive efficiency	η	0.90	-	Accounts for bearings, fit, and minor losses

Table 1. Input values and assumptions used for the preliminary calculation.

2. Basic aerodynamic quantities

Dynamic pressure is calculated from:

$$q = 0.5 \rho V^2$$

At 24 m/s this gives $q = 0.5 \times 1.17 \times 24^2 = 336.96$ Pa. For comparison, at 14 m/s the dynamic pressure is 114.66 Pa. Therefore the aerodynamic loads at 24 m/s are 2.94 times the loads at 14 m/s.

Reynolds number based on the mean wing chord is:

$$Re = V c / \nu$$

$$Re = 24 \times 0.30 / 1.44e-05 = 500000.$$

The area of one aileron is:

$$A_a = b_a c_a$$

$$A_a = 0.7568 \times 0.075 = 0.05676 \text{ m}^2.$$

If a different maximum speed is later used, scale all aerodynamic forces and moments by:

$$(V_{\max} / 24)^2$$

3. Aileron aerodynamic normal force

The preliminary normal-force coefficient is estimated as:

$$C_{N,a} = 2 \pi \tau \delta_a$$

where δ_a is in radians. The force on one aileron is:

$$F_{N,a} = q A_a C_{N,a}$$

Aileron deflection	$C_{N,a}$ lower ($\tau=0.5$)	$C_{N,a}$ conservative ($\tau=1.0$)	Force lower	Force conservative
10 deg	0.548	1.097	10.49 N	20.97 N
15 deg	0.822	1.645	15.73 N	31.46 N
20 deg	1.097	2.193	20.97 N	41.95 N

Table 2. Estimated aerodynamic normal force on one aileron at 24 m/s.

4. Aileron hinge moment and direct-drive torque

The formal hinge-moment expression is:

$$HM_a = c_h q A_a c_a$$

For the current geometry, $q A_a c_a = 336.96 \times 0.05676 \times 0.075 = 1.4344$. Therefore $HM_a = 1.4344 c_h \text{ Nm}$. Because c_h is not known for the exact NACA 2412 aileron geometry, the preliminary force-arm method is also used:

$$HM_a = F_{N,a} x_{cp}$$

with $x_{cp} = 0.4 c_a = 0.030 \text{ m}$.

Aileron deflection	Hinge moment lower	Hinge moment conservative	Direct-drive design torque	Equivalent torque
10 deg	0.315 Nm	0.629 Nm	2.097 Nm	21.4 kg cm
15 deg	0.472 Nm	0.944 Nm	3.146 Nm	32.1 kg cm
20 deg	0.629 Nm	1.258 Nm	4.195 Nm	42.8 kg cm

Table 3. Hinge moment and direct-drive design torque for one aileron at 24 m/s.

For the shown coaxial direct-drive pin concept, there is no servo arm / aileron horn ratio. The servo output torque and the pin torque are therefore approximately equal to the aileron hinge moment before safety factor and losses:

$$T_{\text{pin}} \text{ approximately equals } HM_a$$

The design torque used for pin sizing is:

$$T_{\text{design}} = SF * HM_a / \eta$$

Using the conservative 20 deg case gives $HM_a = 1.258 \text{ Nm}$ and $T_{\text{design}} = 4.195 \text{ Nm}$.

5. Rolling moment from both ailerons

Rolling moment is not the same as servo torque. It is used to check roll-control authority. If the outer edge of the aileron is 0.15 m inboard from the wing tip:

$$y_{\text{outer}} = b/2 - 0.15$$

$$y_{\text{inner}} = y_{\text{outer}} - b_a$$

$$y_{\text{bar}} = (y_{\text{outer}} + y_{\text{inner}}) / 2$$

This gives $y_{\text{outer}} = 1.3650$ m, $y_{\text{inner}} = 0.6082$ m, and $y_{\text{bar}} = 0.9866$ m from the aircraft centerline. The rolling moment from both ailerons is approximated as:

$$L_a = 2 F_{N,a} y_{\text{bar}}$$

Aileron deflection	Rolling moment lower	Rolling moment conservative
10 deg	20.69 Nm	41.39 Nm
15 deg	31.04 Nm	62.08 Nm
20 deg	41.39 Nm	82.77 Nm

Table 4. Approximate rolling moment generated by both ailerons at 24 m/s.

6. Pin torsional sizing

For a solid circular pin in torsion:

$$\tau_{\text{max}} = 16 T / (\pi d^3)$$

Solving for diameter:

$$d_{\text{min}} = [16 T / (\pi \tau_{\text{allow}})]^{1/3}$$

The conservative pin torque is $T = T_{\text{design}} = 4.195$ Nm. The required diameter therefore depends on the allowable shear stress of the pin material:

Allowable shear stress	Required solid pin diameter
50 MPa	7.53 mm
100 MPa	5.98 mm
150 MPa	5.22 mm
200 MPa	4.74 mm

Table 5. Minimum solid pin diameter as a function of allowable shear stress.

The following table shows the torsional shear stress and twist for common pin diameters. The pin length is assumed to be 30 mm.

$$J = \pi d^4 / 32$$

$$\theta = T L / (J G)$$

Pin diameter	Torsional shear stress	Twist, aluminium G=26 GPa	Twist, steel G=79 GPa
3 mm	791.3 MPa	34.87 deg	11.48 deg
4 mm	333.8 MPa	11.03 deg	3.63 deg
5 mm	170.9 MPa	4.52 deg	1.49 deg
6 mm	98.9 MPa	2.18 deg	0.72 deg
8 mm	41.7 MPa	0.69 deg	0.23 deg

Table 6. Pin torsional stress and twist under the conservative 4.195 Nm design torque.

7. Bearing pressure check

If the pin also carries radial load, a simple projected bearing pressure check is:

$$p = F / (d L_b)$$

The conservative 20 deg aerodynamic force is 41.95 N. With safety factor 3, the design radial load is approximately 125.84 N. For a 6 mm pin and one 5 mm long bearing contact:

$$p = 125.84 / (0.006 \times 0.005) = 4.19 \text{ MPa}$$

This pressure is not high for a proper metallic bearing or bushing. For plastic or printed bearing supports, wear, ovalization, alignment, and local crushing become more important than the simple average pressure alone.

8. Mechanical interpretation of the direct-drive pin

The proposed mechanism is a direct rotary drive only if the pin axis is coaxial with the aileron hinge axis and the pin is rigidly locked to the aileron. The load path must be:

servo -> pin -> rigid aileron torque insert -> aileron

The pin may rotate freely inside the wing bearing, but it must not rotate freely relative to the aileron. If the pin is placed inside bearings in both the wing and the aileron without a torque lock to the aileron, the servo will only spin the pin and the aileron will not move.

A round pin should therefore be connected to the aileron by a keyed insert, splined insert, clamp block, bonded torque block, cross-pin, set screw into a flat, or a non-round torque feature. The servo output shaft should not be used as the main structural hinge support. The aileron should still be supported by its own hinge/bearing arrangement so that the servo mainly transmits torque, not bending loads.

9. Design result for the current case

Result	Value
One-aileron area	0.05676 m ²
Dynamic pressure at 24 m/s	336.96 Pa
Reynolds number	500000
Conservative force on one aileron at 20 deg	41.95 N
Conservative hinge moment at 20 deg	1.258 Nm
Direct-drive design torque, SF=3 and eta=0.9	4.195 Nm
Minimum practical solid metal pin diameter	6 mm
Current practical minimum pin length	30 mm
Better stiffness if space allows	8 mm diameter x 30 mm length

Table 7. Summary of the preliminary load and pin-sizing result.

Final recommendation for the pin, without selecting a servo: For the current assumptions, the calculated practical minimum is a solid metal pin of 6 mm diameter and 30 mm length. A 6 mm pin gives approximately 98.9 MPa torsional shear stress and about 0.72 deg twist if steel is used. If space is available, an 8 mm pin is mechanically better because it reduces the shear stress to about 41.7 MPa and the steel twist to about 0.23 deg.

Design caution. The value 6 mm is not a universal absolute minimum. It depends on material, manufacturing quality, torque-lock design, bearing support, alignment, and the real maximum flight speed. If the aircraft can fly faster than 24 m/s, all aerodynamic forces, hinge moments, and direct-drive torques must be increased by $(V_{\max}/24)^2$.

References

Holzapfel, F. Fundamentals of Flight Control, lecture slides, Institute of Flight System Dynamics, Technical University of Munich. Used for the aileron/control-surface and actuator context.

Scholz, D. TN-EV52-360/91, Aircraft Design - Hinge Moments and Actuator Sizing. Used for the hinge-moment and actuator-sizing logic $HM = c_h q A c$.